

Problem Set 7 Answers

1a. We note that $X = x$ if and only if Bob gets $x - 1$ non-Aces in a row, followed by an Ace, which yields $P(X = x) = (48/52)^{x-1}(4/52) = (12/13)^{x-1}(1/13)$.

1b. Similarly, we have $X \geq x$ if and only if Bob's first $x - 1$ draws are not Aces, which yields $P(X \geq x) = (48/52)^{x-1} = (12/13)^{x-1}$.

2. To have exactly x heads, there are $\binom{100}{x}$ ways that x out of 100 flips will be heads, so $P(X = x) = \binom{100}{x}(1/2)^x(1/2)^{100-x} = \binom{100}{x}(1/2)^{100}$.

3. To have exactly x green bears, there are $\binom{3}{x}$ ways that x out of the 3 bears will be green, so $P(X = x) = \binom{3}{x}(1/6)^x(5/6)^{3-x}$.

4. We have:

$$P(X = 1) = 3/10$$

$$P(X = 2) = (7/10)(3/9) = 7/30$$

$$P(X = 3) = (7/10)(6/9)(3/8) = 7/40$$

$$P(X = 4) = (7/10)(6/9)(5/8)(3/7) = 1/8$$

$$P(X = 5) = (7/10)(6/9)(5/8)(4/7)(3/6) = 1/12$$

$$P(X = 6) = (7/10)(6/9)(5/8)(4/7)(3/6)(3/5) = 1/20$$

$$P(X = 7) = (7/10)(6/9)(5/8)(4/7)(3/6)(2/5)(3/4) = 1/40$$

$$P(X = 8) = (7/10)(6/9)(5/8)(4/7)(3/6)(2/5)(1/4)(3/3) = 1/120$$

If we use the shorthand $a^{\underline{j}} = (a)(a-1)(a-2) \cdots (a-j+1)$ as the j th falling power of a , then we can simplify this formula to:

$$P(X = x) = \frac{7^{\underline{x-1}}}{10^{\underline{x-1}}} \cdot \frac{3}{10 - x + 1}$$

5. The probability that Bob gets 1 head is: $(1/6)\binom{1}{1}(1/2)^1 + (1/6)\binom{2}{1}(1/2)^2 + (1/6)\binom{3}{1}(1/2)^3 + (1/6)\binom{4}{1}(1/2)^4 + (1/6)\binom{5}{1}(1/2)^5 + (1/6)\binom{6}{1}(1/2)^6 = \sum_{y=1}^6 (1/6)\binom{y}{1}(1/2)^y = 5/16$. It follows that, for $1 \leq y \leq 6$, the value of $P(Y = y \mid X = 1)$ is exactly

$$\frac{(1/6)\binom{y}{1}(1/2)^y}{(1/6)\binom{1}{1}(1/2)^1 + (1/6)\binom{2}{1}(1/2)^2 + (1/6)\binom{3}{1}(1/2)^3 + (1/6)\binom{4}{1}(1/2)^4 + (1/6)\binom{5}{1}(1/2)^5 + (1/6)\binom{6}{1}(1/2)^6}$$

which simplifies to:

$$P(Y = y \mid X = 1) = \left(\frac{8}{15}\right)\binom{y}{1}(1/2)^y.$$

For y from 1 to 6, these values are, respectively, $4/15$, $4/15$, $1/5$, $2/15$, $1/12$, $1/20$.